

Resit Exam – Statistical Data Analysis

July 2, 2024

Exam Info

- The use of a basic calculator is allowed
- Graphical calculators and mobile phones are *not* allowed
- Please write all of your answers in English
- This exam consists of 5 questions on 5 pages
- This exam is worth a total of 45 points
- Denoting by τ the number of achieved points, the exam grade is $\frac{\tau+5}{5}$
- You have 2 hours and 45 minutes to write the exam

Good Luck 😊

Question 1 [10 points]

Mark each of the following statements as *True* or *False* and shortly motivate your answers (even for the statements you deem to be true). *Note*: a correct assessment without explanation will grant you 0.5 points.

- [2 points] By choosing an extremely large number of bootstrap samples, any error arising in the implementation of a bootstrap procedure can be made arbitrarily small.
- [2 points] In kernel density estimation, one can only use the normal kernel if the distribution underlying the data appears to be the normal distribution.
- [2 points] In some situations, the (Wilcoxon) signed rank test is applicable while the one-sample t -test for location is not.
- [2 points] The term *multiple* linear regression refers to the fact that the response variable is multivariate.
- [2 points] The coefficient of determination $\mathcal{R}^2$ should *not* be used to compare linear models with a different number of explanatory variables.

Question 2 [8 points]

In this exercise, we cover several topics.

- [2 points] Explain the difference in the null hypotheses that can be tested with the Shapiro-Wilk test and the (one-sample) Kolmogorov-Smirnov test.
- [2 points] In the context of kernel density estimation, explain why the choice of bandwidth is more important than the choice of kernel.
- [2 points] In the context of linear regression, explain why a point with high leverage is not necessarily an influence point.
- [2 points] In the context of linear regression, explain how a collinearity issue impacts model fit.

Question 3 [10 points]

Let a sample $\mathbf{x}$ consist of realizations $x_1, \dots, x_n$ of i.i.d. random variables $X_1, \dots, X_n$ from a distribution F . Figure 1 below consists of a histogram depicting sample $\mathbf{x}$ and of two QQ -plots. Here, we are interested in the statistic

$$T_n(X_1, \dots, X_n) = \text{med}(X_1, \dots, X_n) - \bar{X},$$

where $\text{med}(X_1, \dots, X_n)$ denotes the sample median and $\bar{X}$ the sample mean. This statistic can be used as a measure of skewness.

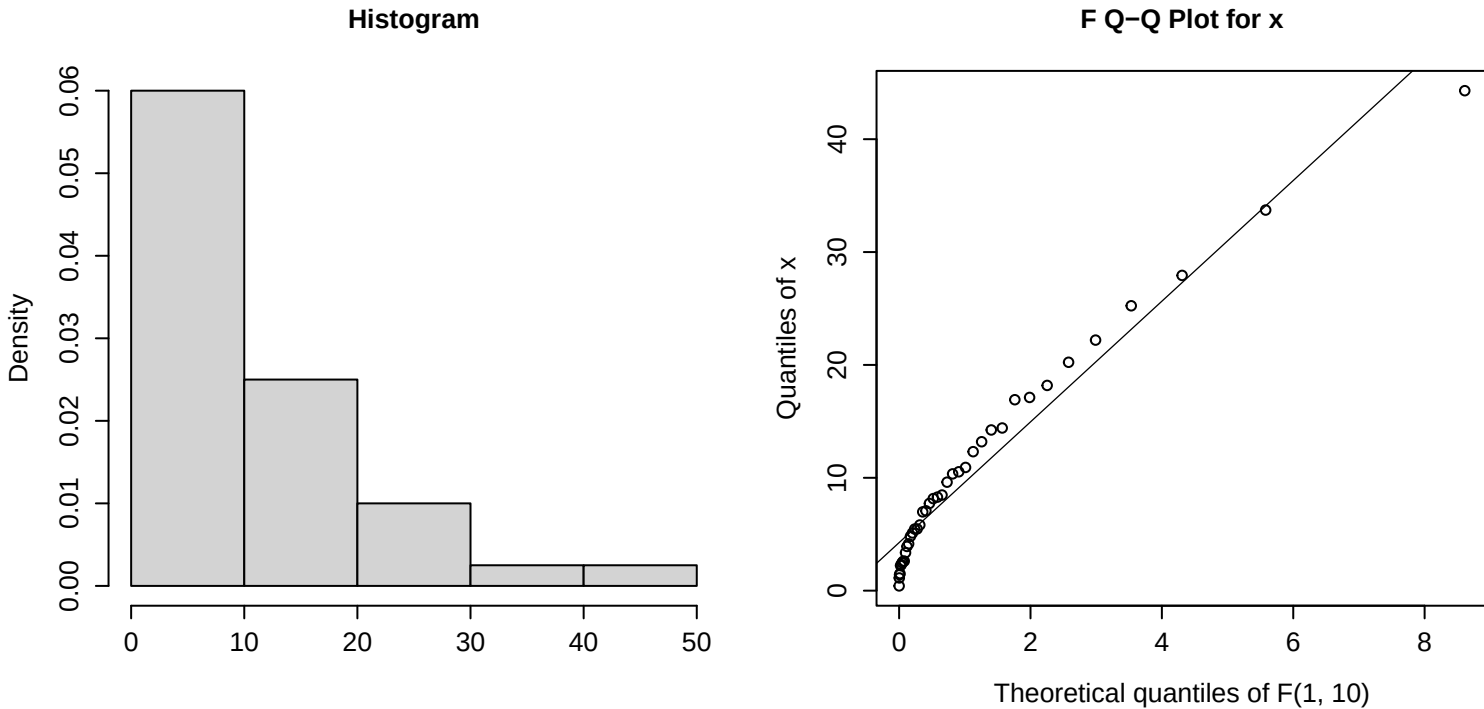


Figure 1: Histogram of sample $\mathbf{x}$ (leftmost panel) and two QQ -plots.

- [2 points] Based on the plots in Figure 1, which location-scale family do you think is the most appropriate for these data and why?
- [2 points] Can the (Wilcoxon) signed rank test be used to test hypotheses about the median of the distribution underlying $\mathbf{x}$ or are any of its assumptions violated? Motivate your answer.
- [2 points] The statistic computed from sample $\mathbf{x}$ is negative, i.e. $T_n(\mathbf{x}) = -2.63$. Does this indicate that the distribution of $\mathbf{x}$ is rather left-skewed, right-skewed or symmetric? Why?
- [3 points] We wish to use a *parametric* bootstrap to approximate the distribution of the statistic T_n . Describe the steps of a reasonable parametric bootstrap scheme for estimating the standard deviation of T_n . In particular, what is necessary in order to perform the sampling step of this parametric bootstrap?
- [1 point] If we use the *empirical* bootstrap instead, how do we generate bootstrap samples?

Question 4 [8 points]

A sample of $n = 120$ students taking a statistics course were asked if they spend their holidays in Italy and if they put ketchup on pizza. The results are displayed in Table 1 below.

	YES Ketchup on 🍕	NO Ketchup on 🍕	TOT
YES Holidays in 🇮🇹	4	36	40
NO Holidays in 🇮🇹	26	54	80
TOT	30	90	120

Table 1: Cell frequencies (counts) for vacationing in Italy-using ketchup on pizza combinations.

- a. [2 points] Given these data and how they were collected, would one test a null hypothesis of *independence* between vacationing in Italy and using ketchup on pizza or a null hypothesis of *homogeneity*? Justify your answer.
- b. [3 points] The corresponding test is based on the test statistic

$$X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(N_{ij} - n\hat{p}_{ij})^2}{n\hat{p}_{ij}}$$

where r denotes the number of rows, c the number of columns, N_{ij} the observed cell frequency (count) for cell (i, j) and $\hat{p}_{ij}$ the corresponding estimated null probability.

Under the null hypothesis, the test statistic X^2 approximately follows a $\chi^2_{(r-1)(c-1)}$ distribution if a certain rule of thumb is satisfied. Explain what this rule of thumb is, and check whether it is satisfied for the present data.

- c. [2 points] For the data in Table 1, we have $X^2 = 6.05$, to be compared with the appropriate critical value in Table 2 below (significance level: $\alpha = 5\%$). Do you reject the null hypothesis of the test or not? Justify your answer.

k	$p = 0.025$	$p = 0.05$	$p = 0.5$	$p = 0.95$	$p = 0.975$
1	0.001	0.004	0.455	3.841	5.024
2	0.051	0.103	1.386	5.991	7.378
3	0.216	0.352	2.366	7.815	9.348
4	0.484	0.711	3.357	9.488	11.143
5	0.831	1.145	4.351	11.070	12.833
6	1.237	1.635	5.348	12.592	14.449
7	1.690	2.167	6.346	14.067	16.013
8	2.180	2.733	7.344	15.507	17.535
9	2.700	3.325	8.343	16.919	19.023
10	3.247	3.940	9.342	18.307	20.483

Table 2: p -quantiles (rounded to the third decimal) of the χ^2_k -distribution for given degrees of freedom k .

- d. [1 point] Another test, which is sometimes applicable in the context of contingency tables, is Fisher's exact test. Can this test be applied in this situation? Why/Why not?

Question 5 [9 points]

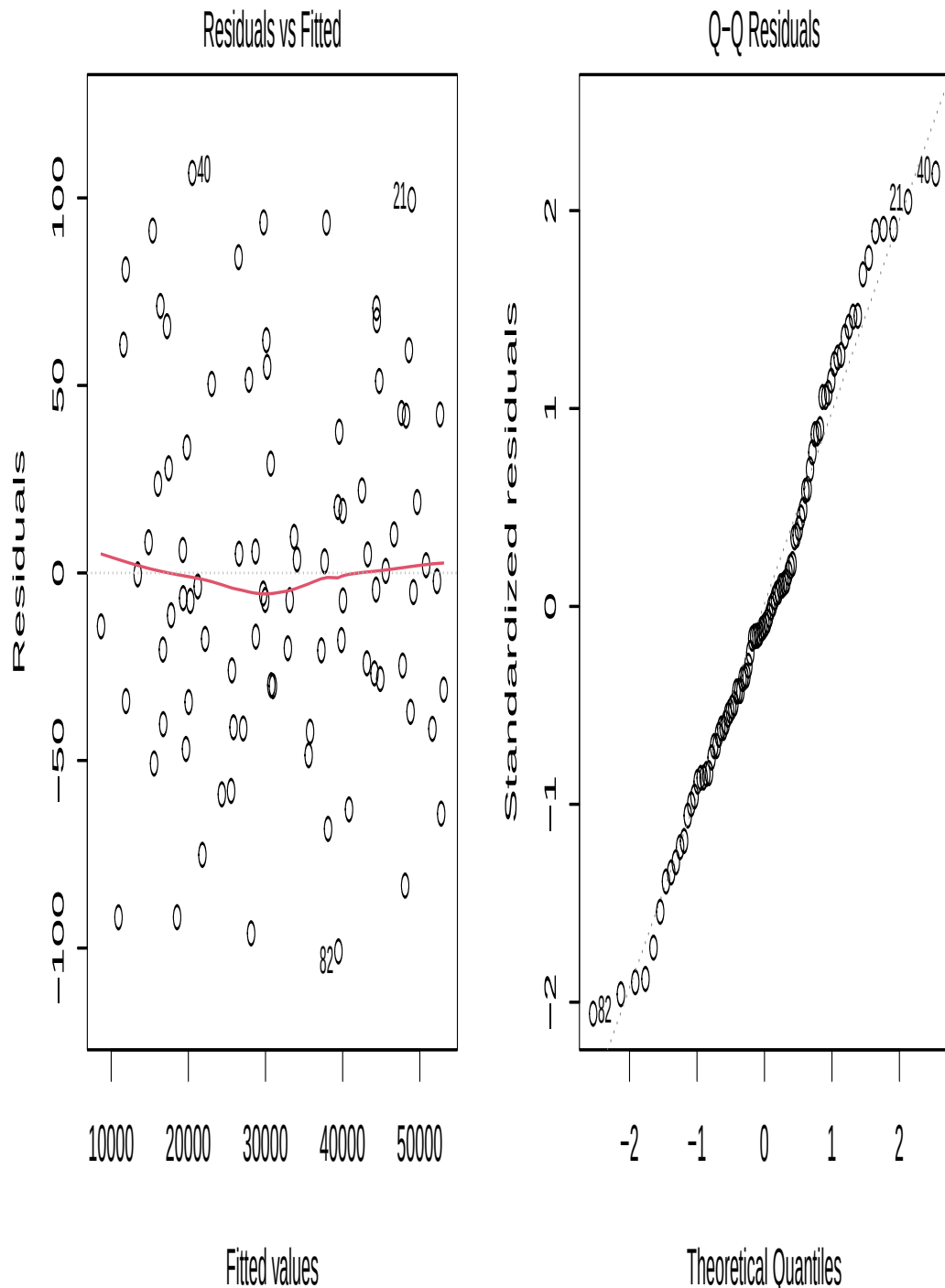
You analyze data of $n = 90$ prices of diamonds and fit the linear model: $\text{price} = \beta_0 + \beta_1 \cdot \text{weight} + \beta_2 \cdot \text{clarity} + \beta_3 \cdot \text{depth} + \epsilon$, where **weight** is the weight of the stone (in carats), **clarity** is a measurement of how clear the stone is (on a 1 – 100 scale) and **depth** is the depth of the stone (in mm). The estimates given by R are as follows:

```
##           Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  309.97883   84.081596    3.686643  3.969820e-04
## weight      9282.25242    4.602283  2016.880138  8.203209e-203
## clarity     123.78424    1.219761   101.482373  2.592606e-91
## depth      -97.70985     0.822064  -118.859174  3.564518e-97
```

- a. [1 points] Is the coefficient β_3 significantly different from zero (significance level: $\alpha = 5\%$)? Briefly justify your answer.
- b. [2 points] According to the fitted linear model, what is the estimated price of a diamond with a **weight** of 1 carat, a **clarity** of 100 and a **depth** of 10mm?

Hint: first, write down the fitted model using the estimated coefficients. There is no need to consider the error term, as in expectation it is zero.

- c. [2 points] Describe two assumptions of the linear model that can be checked with plots like the ones below, and how to do so.



- d. [2 points] The variance inflation factors (VIFs) corresponding to the fitted model are 1.493, 1.035 and 1.471 , respectively. Based on these numbers, discuss whether there seems to be a collinearity problem with the model (justify your answer).
- e. [2 points] The left panel of the plot below shows the values of Cook's distance D_i (on the y -axis) for all observations (the observation indices $i = 1, \dots, n$ are on the x -axis). The right panel is the **Residual vs Leverage** plot, with contour lines (dashed) corresponding to Cook's distance values of 0.1 and 0.15 added. Are there any influence points in the model? Briefly justify your answer (you can use either plot to do so, it is not required to comment on both).

